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Research Article

Autoactivation of mycorrhizal symbiosis signaling through gibberellin deactivation in orchid seed germination

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Abstract

Orchids parasitically depend on external nutrients from mycorrhizal fungi for seed germination. Previous findings suggest that orchids utilize a genetic system of mutualistic arbuscular mycorrhizal (AM) symbiosis, in which the plant hormone gibberellin (GA) negatively affects fungal colonization and development, to establish parasitic symbiosis. Although GA generally promotes seed germination in photosynthetic plants, previous studies have reported low sensitivity of GA in seed germination of mycoheterotrophic orchids where mycorrhizal symbiosis occurs concurrently. To elucidate the connecting mechanisms of orchid seed germination and mycorrhizal symbiosis at the molecular level, we investigated the effect of GA on a hyacinth orchid (*Bletilla striata*) seed germination and mycorrhizal symbiosis using asymbiotic and symbiotic germination methods. Additionally, we compared the transcriptome profiles between asymbiotically and symbiotically germinated seeds. Exogenous GA negatively affected seed germination and fungal colonization, and endogenous bioactive GA was actively converted to the inactive form during seed germination. Transcriptome analysis showed that *B. striata* shared many of the induced genes between asymbiotically and symbiotically germinated seeds, including GA metabolism- and signaling-related genes and AM-specific marker homologs. Our study suggests that orchids have evolved in a manner that they do not use bioactive GA as a positive regulator of seed germination and instead autoactivate the mycorrhizal symbiosis pathway through GA inactivation to accept the fungal partner immediately during seed germination.

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Introduction

Seed germination is an important process in the plant life cycle because it determines subsequent plant survival and reproductive success (Rajjou et al. 2012). The plant hormone gibberellin (GA) plays an essential role in promoting seed germination in many plant species (Shu et al. 2016). During seed imbibition of photosynthetic plants with endosperm, such as cereal crops, the embryo synthesizes GA, which is subsequently released to aleurone cells to activate the synthesis and secretion of hydrolases such as α-amylase (Tuan et al. 2018). These hydrolases degrade starch and other nutrients stored in the endosperm into small molecules that are used by embryos for supporting seed germination and seedling growth (Tuan et al. 2018). GA also stimulates the germination of exalbuminous seeds, including Brassicaceae, Asteraceae, Solanaceae, and Fabaceae (Toyomasu et al. 1993; Leubner-Metzger 2001; Ogawa et al. 2003; Sheokand et al. 2005), which store a large amount of lipid and protein content in their cotyledons. It has been suggested that GA plays a role in weakening the mechanical barrier of the micropylar endosperm for a prerequisite of radicle protrusion (Finch-Savage and Leubner-Metzger 2006).

Orchids, which belong to the family Orchidaceae, produce a large number of small seeds with no endosperm, which are termed dust-like seeds (Arditti and Ghani 2000). The mature seeds, consisting only of a globular embryo surrounded by a thin seed coat, contain few or no starch granules (Yeung 2017). Lee et al. (2008) found that lipid and protein bodies are the main storage products in moth orchid (*Phalaenopsis amabilis*) embryos. Under natural conditions, all orchid seeds parasitically rely on orchid mycorrhizal (OM) fungi to obtain carbon, nitrogen, and phosphorus during the early developmental stage (Leake 1994), which is called "initial mycoheterotrophy" (Merckx 2013).

OM fungi penetrate seeds through the suspensors (Peterson and Currah 1990; Richardson et al. 1992; Rasmussen and Rasmussen 2009) and subsequently through rhizoids (Williamson and Hadley 1970) and form dense coils of mycelium called pelotons in the cortical cells at the suspensor side of the embryo. Orchids obtain nutrients through degraded (Bougoure et al. 2014) or both live and degenerating pelotons (Kuga et al. 2014). Most orchids form symbioses with Rhizoctonia-like fungi, which are free living and saprophytic, belonging to the phylum Basidiomycota (Rasmussen 2002; Yukawa et al. 2009). Orchidaceae is sister to the families of the Asparagales, which, unlike orchids, form a symbiosis with Glomeromycotina fungi. This association, termed arbuscular mycorrhizae (AM), is regarded as the most ancient plant–fungus interaction (Delaux et al. 2013). It has been speculated that the ancestors of orchids formed AM (Yukawa et al. 2009; Selosse et al. 2022), although it is yet to be elucidated (Rasmussen and Rasmussen 2014). Consistent with this, the common symbiotic genes involved in a putative signal transduction pathway shared by AM and the rhizobium–legume symbiosis, such as symbiosis receptor-like kinase *SymRK*, calcium- and calmodulin-dependent protein kinase *CCaMK*, and calcium signal decoding protein *CYCLOPS*, are present in many orchid species: *Apostasia shenzhenica*, *Dendrobium catenatum*, *Phalaenopsis equestris*, and *Gastrodia elata* (Radhakrishnan et al. 2020; Xu et al. 2021). These gene components are absent in other non-AM plants, such as *Arabidopsis* (*Arabidopsis thaliana*, Brassicaceae), loblolly pine (*Pinus taeda*, Pinaceae), and red water fern (*Azolla filiculoides*, Salviniaceae) (Radhakrishnan et al. 2020). Our previous molecular-based study of a hyacinth orchid (*Bletilla striata*) has shown that orchids share molecular components such as common symbiotic genes and their concomitant expression common to AM plants to establish mycoheterotrophic symbiosis. These findings suggest that orchids indeed may utilize a genetic system of AM symbiosis inherited from their common ancestor (Miura et al. 2018).

In terms of AM symbiosis, recent studies have revealed that abnormal elevation of GA signaling decreases the colonization of the host root by AM fungi, and exogenous GA reduces hyphal colonization and arbuscule formation during AM symbiosis (Floss et al. 2013; Foo et al. 2013; Takeda et al. 2015). The arbuscule formation requires the presence of DELLA (aspartic acid–glutamic acid–leucine–leucine–alanine) proteins, which are negative regulators of GA signaling (Floss et al. 2013). In a model legume *Lotus japonicus*, a complex comprising CCaMK, CYCLOPS, and DELLA binds to the promoter of *REDUCED ARBUSCULAR MYCORRHIZA1*, a central regulator of arbuscule development (Pimprikar et al. 2016). Thus, GA signaling negatively affects AM fungal colonization and development.

More than 70% of the angiosperm species establish AM symbiosis (Brundrett 2009). The autotrophic AM plants form root symbiosis after germination using nutrient sources stored in their seeds. This means that the mechanisms of seed germination and mycorrhizal symbiosis are independent of most autotrophic plants. In contrast, these 2 events, for both of which GA is a negative or positive key regulator, occur concurrently in orchids. Previous studies showed that GA had negative or no significant effect on symbiotically and asymbiotically germinated seedlings of phylogenetically distant terrestrial and epiphytic orchids (Hadley and Harvais 1968; Van Waes and Debergh 1986; Wilkinson et al. 1994; Miyoshi and Mii 1995; Chen et al. 2020). Another report showed the positive effect of GA on asymbiotically germinated immature seeds of a small white orchid (*Pseudorchis albida*), probably due to the differences in the degree of seed maturity (Pierce and Cerabolini 2011). Because germinating orchid seeds obtain nutrients from an exogenous source (i.e. OM fungi) (Arditti and Ghani 2000), the lack of internal endosperm may explain low sensitivities to GA. Previous orchid transcriptomic studies have reported high expression of genes related to GA biosynthesis, catabolism, and signaling during symbiotic germination (Zhao et al. 2014; Liu et al. 2015; Miura et al. 2018). Chen et al. (2020) showed inhibition of fungal colonization in symbiotically germinated seeds when applying high concentrations of exogenously GA. Given these findings, we hypothesized that the fact

that orchids have evolved to not use GA as a positive regulator of seed germination contributes to establishing and maintaining symbiotic associations concurrently with seed germination. To elucidate the connecting mechanisms of orchid seed germination and mycorrhizal symbiosis at the molecular level, we investigated the effect of GA on seed germination and mycorrhizal symbiosis by using a terrestrial orchid, *B. striata* cv. Murasaki Shikibu (subfamily Epidendroideae, tribe Arethuseae), which is autotrophic at adulthood. The features that *B. striata* can grow fast among orchids and almost synchronously in vitro both asymbiotically and symbiotically make this species a potentially useful model species for OM research (Yamamoto et al. 2017). The results of asymbiotic and symbiotic germination tests with and without GA showed that GA negatively affected both seed germination and fungal colonization. The GA-treated asymbiotic germination assay confirmed the negative effects on some phylogenetically distant orchids. Asymbiotically and symbiotically germinated *B. striata* protocorms were subjected to transcriptome analysis using RNA-sequencing (RNA-seq) to determine whether the germination procedures share differentially expressed genes (DEGs) between the 2 conditions. The results showed that the asymbiotically and symbiotically germinated protocorms (hereafter APs or SPs, respectively) shared more than half of the total DEGs. The AM-specific marker homologs were upregulated not only in SPs but also in APs, meaning that symbiosis signaling is activated during seed germination even without fungi. During seed germination, the expression of GA 2-oxidase-encoding genes, encoding enzymes that convert bioactive GAs to inactive forms, was drastically increased, and endogenous bioactive GA was actively converted to the inactive form. Our study, therefore, concluded that orchids autoactivate the mycorrhizal symbiosis pathway during seed germination through GA inactivation, suggesting an adaptive mechanism to reconcile the 2 events, seed germination and mycorrhizal symbiosis.

Results

GA negatively affects both fungal colonization and seed germination in *B. striata*

Surface-sterilized *B. striata* seeds were sown on an oatmeal agar (OMA) medium containing 1 μM GA and inoculated with the previously isolated *Tulasnella* sp. HR1-1 (hereafter *Tulasnella*) (Yamamoto et al. 2017) to examine the effects of GA on symbiosis. No significant effect was found on the mycelial growth of GA-treated *Tulasnella* (Supplemental Fig. S1A). Since the protocorm colonized with *Tulasnella* contains dozens of symbiotic cells with hyphal coils (Fig. 1A), the symbiotic efficiency between *B. striata* and *Tulasnella* was evaluated by comparing the number of symbiotic cells within a protocorm. A 2-wk-old protocorm contained approximately 81.0 ± 4.0 symbiotic cells with hyphal coils in the control treatment (Fig. 1B). The number of symbiotic cells in 1 μM GA treatment was significantly lower (5.4 ± 3.0) than in the control (Fig. 1B), while no differences were detected in terms of fungal entry at the suspensor end between the treatments (Supplemental Fig. S1B). On the contrary, a GA biosynthesis inhibitor uniconazole-P treatment showed a significant increase in symbiotic cells (135.8 ± 38.2, Fig. 1B). Uniconazole-P exhibited a positive effect on hyphal growth, but it was not significant compared with GA treatment (Supplemental Fig. S1A). These results resembled those of earlier studies on AM symbiosis in which exogenous GA reduced hyphal colonization and arbuscule formation (El Ghachtouli et al. 1996; Yu et al. 2014; Takeda et al. 2015). The exogenous GA also significantly inhibited seed germination of *B. striata* inoculated with *Tulasnella* (Fig. 1C), in line with previous reports (Chen et al. 2020).

Our findings support previous studies that OM symbiosis may share at least some common properties with AM symbiosis (Suetsugu et al. 2017; Miura et al. 2018) but do not support the general notion that GA stimulates seed germination. The *B. striata* seeds were then sown asymbiotically on HYPONeX-sucrose agar (HA) medium with or without GA. The results of germination rates demonstrated that 1 and 10 μM GA treatment, as well as 1 μM abscisic acid (ABA), significantly suppressed germination (25.2 ± 3.6% and 20.2 ± 3.8% in 1 and 10 μM GA, respectively) as compared with a control experiment (65.9 ± 8.6%), while 0.1 μM GA showed no significant effect (55.4 ± 4.4%) (Fig. 2A; Supplemental Fig. S1C). The treatment of 0.1 and 1 μM uniconazole-P showed a promotion effect on seed germination (88.6 ± 4.1% and 83.2 ± 5.4% in 0.1 and 1 μM uniconazole-P, respectively), while 10 μM of uniconazole-P negatively affected the germination (45.0 ± 9.7%) (Fig. 2A). Because uniconazole-P has also been reported to inhibit the biosynthesis of other phytohormones such as ABA and brassinosteroids (Iwasaki and Shibaok 1991; Kitahata et al. 2005; Saito et al. 2006), a simultaneous treatment test with GA and uniconazole-P was conducted. Asymbiotically germinated seeds that were simultaneously treated with 1 μM GA and 1 μM uniconazole-P showed a lower germination rate than the control as well as with only GA treatment (Fig. 2B). Moreover, germination was significantly greater when *B. striata* seeds were grown on filter paper saturated with 1 μM uniconazole-P solution without any nutrition (Fig. 2C). The negative GA effect or positive uniconazole-P effect was not found in the seed germination of *Oryza sativa* and *L. japonicus* (Supplemental Fig. S2).

To test whether other orchids show a similar pattern as GA-treated *B. striata* seeds, an asymbiotic germination experiment was conducted using seeds of 4 other orchid species: 3 terrestrial (austral ladies tresses [*Spiranthes australis*], *Goodyera crassifolia*, and *Cremastra appendiculata* var. *variabilis*), and 1 epiphytic (wind orchid [*Vanda falcata* cv. Beniohgi]). Similar to the results for *B. striata*, exogenous GA treatment decreased the germination percentage of these orchids (Table 1). In addition, commercial plant growth regulators containing substances that inhibit GA biosynthesis showed activity as seed germination stimulators (Supplemental Fig. S1D).

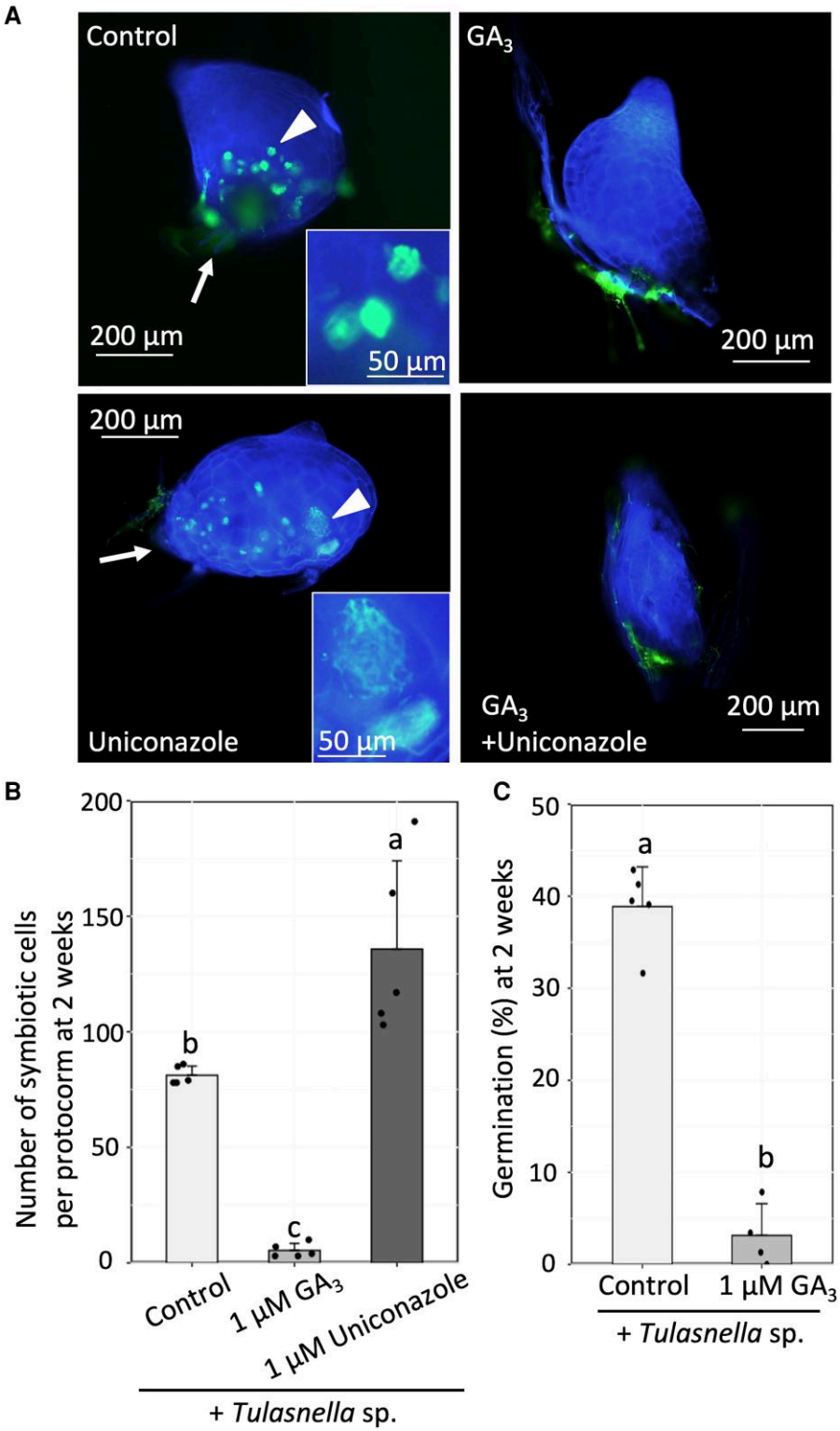

Figure 1. Effects of GA (GA_3) and its inhibitor (uniconazole-P) on SPs. **A)** Symbiotically germinated *B. striata* seeds at 2 wk after sowing (WAS) on OMA medium inoculated with *Tulasnella* sp. HR1-1. The images show fungal-infected protocorms stained with calcofluor white (blue) and wheat germ agglutinin-Alexa fluor-488 (green) to visualize the plant cell and fungal structures, respectively. Green fluorescent dots at the suspensor side (arrowheads) and the inset indicate fungal pelotons. Arrows indicate the suspensor end. **B, C**) The number of symbiotic cells **B)** and germination percentages **C)** at 2WAS on OMA medium inoculated with *Tulasnella* sp. HR1-1. Symbiotic cells and germinated seeds treated with 0.01% (v/v) ethanol for the control, 1 μM GA_3, or 1 μM uniconazole-P were counted. Different letters in **B)** indicate significant differences among treatments on the basis of a Bonferroni-adjusted pairwise *t*-test ($n = 5$ individual experiments, each containing 5 protocorms, $P < 0.05$). *P*-value was calculated on the basis of Student's *t*-test ($n = 4$ to 5 replicate plates, each containing 185 ± 64 seeds, $P < 0.05$) **C)**. Each bar represents the mean value $\pm$ SD. All experiments were independently performed 3 times with similar results.

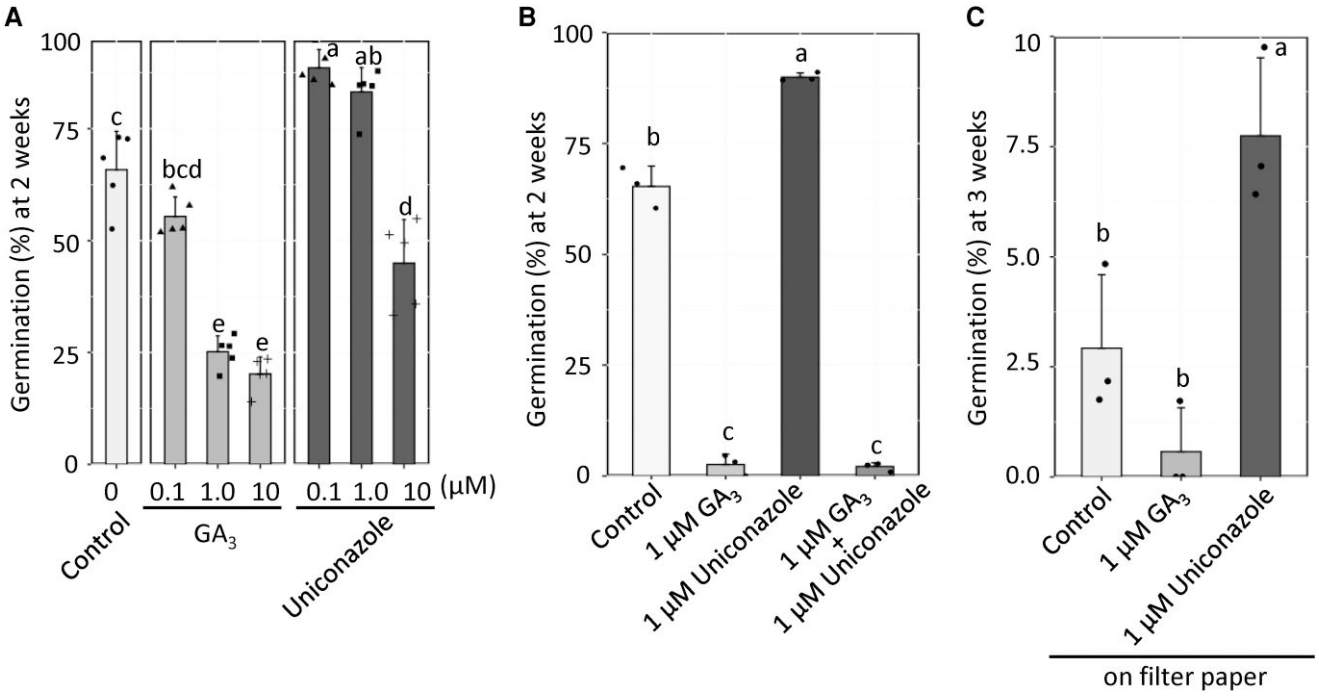

Figure 2. Effects of GA (GA_3) and its inhibitor (uniconazole-P) on APs. *B. striata* seeds were sown asymbiotically on Hyponex agar medium **A, B)** or filter papers **C)**. Germinated seeds treated with 0.01% (v/v) ethanol for the control, GA_3, or uniconazole-P at the indicated concentrations were counted at 2 **A, B)** or 3 **C)** wk after sowing. Different letters indicate statistically significant differences on the basis of a Bonferroni-adjusted pairwise *t*-test ($n = 3$ to 5 replicate plates, each containing 117 ± 54 seeds, $P < 0.05$). Each bar represents the mean value ± SD. All experiments were independently performed 3 times with similar results.

Table 1. The effect of GA on orchid seed germination

Species name	Period	Control (%)[a]	GA (%)[a]	Uniconazole-P (%)[a]
Spiranthes australis	28 d	4.90 ± 1.36^b	1.09 ± 0.49^c	5.64 ± 0.57^b
Goodyera crassifolia	28 d	10.45 ± 2.75^b	3.79 ± 1.52^b	18.34 ± 11.72^b
Cremastra appendiculata var. *variabilis*	60 d	5.04 ± 1.26	Not detected	7.22 ± 4.50
Vanda falcata cv. *Beniohgi*	28 d	51.30 ± 4.85^b	29.89 ± 4.00^c	Not tested

[a]Numbers percentages indicate the germination percentages.
[b,c]Different characters indicate statistically significant differences on the basis of the Bonferroni-adjusted pairwise *t*-test ($n = 3$ to 5, $P < 0.05$). The germination percentage represents the mean value ± SD.

These results revealed that exogenous GA negatively affected seed germination of phylogenetically distant orchids, indicating that seed germination and fungal colonization are promoted through the inhibition of de novo GA biosynthesis.

A majority of DEGs were shared between APs and SPs

The negative effects of GA on both seed germination and fungal colonization in *B. striata* led us to hypothesize that orchid germination mechanisms are linked with the symbiotic system. To gain insight into the similarity and differences between asymbiotic and symbiotic germination mechanisms at a molecular level, we performed a time-course transcriptome profiling on APs or SPs (Supplemental Table S1). Our previous study showed that *B. striata* grew almost synchronously for the first 3 wk with both the germination methods, asymbiotic germination on the HA medium and symbiotic germination with *Tulasnella* on the OMA medium (Yamamoto et al. 2017). In addition, symbiotic protocorms were dominated by fungal pelotons at the early (initiation of hyphal coiling), middle (well-developed hyphal coils), and late (hyphal degradation) stages within 1, 2, and 3 wk after sowing, respectively (Yamamoto et al. 2017; Miura et al. 2018). Upregulated and downregulated DEGs were identified from comparisons of each week-old (Weeks 1, 2, and 3) protocorms versus seeds at the start of the experiment (Week 0) with a $|\text{Log}_2\text{-fold change (FC)}|$ threshold of ≥ 1 and a false discovery rate (FDR) threshold of <0.05 (Supplemental Table S2). A total of approximately 10,000 to 15,000 genes in Week 1 to 3 protocorms had a significantly higher expression compared with Week 0 seeds. Among these DEGs, 8,217 to 8,693 genes were common in both APs and SPs comparison sets, which covered 38.6% to 52.5% of the total number of DEGs from Weeks 1 to 3 (Fig. 3A). A total of approximately 6,000 genes in Week 1 to 3 protocorms had significantly lower expression than Week 0 seeds (Fig. 3A). Among these, around 4,500 genes were common

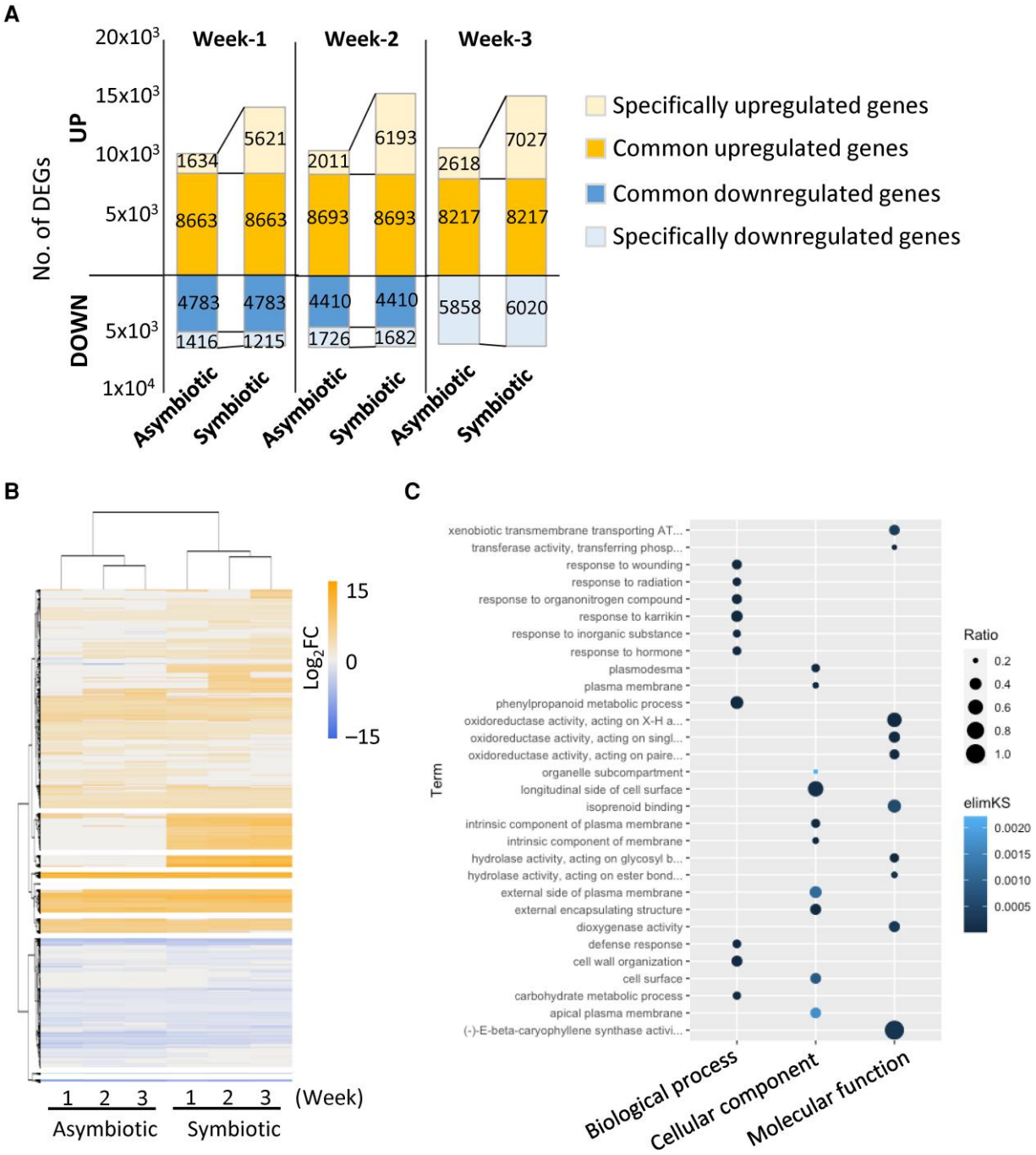

Figure 3. Transcriptome analysis of asymbiotically and symbiotically germinated *B. striata*. **A)** The bar chart of the number of DEGs. Gene expression levels of APs (asymbiotic) and SPs (symbiotic) at 1 to 3 wk after seeding were compared with Week 0 seeds. **B)** Hierarchical clustering of DEGs. The method of k-means clustering was used to identify similarities in expression patterns among asymbiotic and symbiotic. The heatmap was drawn by the pheatmap package in R. **C)** GO enrichment analysis of shared overexpressed genes between asymbiotic and symbiotic at Week 1. The most significant 10 terms of each category, biological process, cellular component, and molecular function, are shown on the basis of the elim-Kolmogorov–Smirnov (elimKS) method in the topGO package in R (elimKS < 0.01). All significant terms were presented in Supplemental Table S4.

between SP and AP in Weeks 1 and 2, and no common genes were detected in Week 3 (Fig. 3A).

To compare the transcription profiles of seed germination and mycorrhizal symbiosis in other plant species, DEGs were identified using the previous transcriptome data of seed germination on Day 0 versus Day 2 and AM roots versus non-AM roots of rice (*O. sativa*) (Narsai et al. 2017; Kobae et al. 2018). The rice transcriptome data showed that shared DEGs between seed germination and AM symbiosis covered less than 24% of the total DEGs (Supplemental Fig. S3A). This

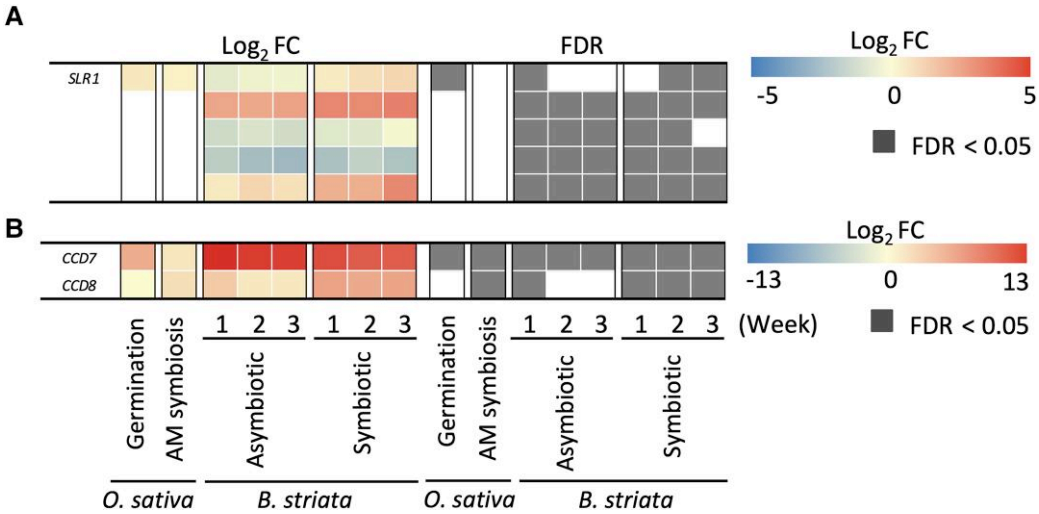

Figure 4. The expression patterns of GA signaling and SL biosynthesis genes identified during RNA-seq analyses. **A)** The expression patterns of *SLENDER RICE1* (*SLR1*) putative orthologs containing DELLA domain. **B)** The expression patterns of carotenoid cleavage dioxygenase genes (*CCD7* and *CCD8*) involved in SL biosynthesis. The heatmap on the left shows the expression patterns of the selected genes on the basis of Log_2FC. Log_2FC was calculated between time points; 0-wk seeds versus 1- to 3-wk protocorms. "Germination" and "AM symbiosis" indicate germinated seeds and arbuscular mycorrhizal roots of *O. sativa*, respectively. The right panel displays FDRs.

is a much lower percentage than the *B. striata* common DEGs.

Hierarchical clustering based on the *B. striata* expression profiles showed that the transcripts were divided into AP and SP, as expected (Fig. 3B). Transcripts at Week 1 exhibited different expression patterns from those of Weeks 2 and 3 (Fig. 3B). Additionally, over 50% of DEGs were shared between AP and SP at Week 1 (Fig. 3A), and GA restricted peloton formation but still allowed the fungal entry at the suspensor end (Supplemental Fig. S1A). On the basis of these results, it was expected that the stage of forming a shoot apex in APs or fungal colonization following fungal entry from the suspensor end to cortical cells in SPs is the key to understanding mycoheterotrophic germination. Then, the shared and specific DEGs at Week 1 were annotated into 3 categories ("biological process," "cellular component," and "molecular function") after gene ontology (GO) enrichment analysis with a cutoff *P*-value of 0.01 (Supplemental Table S3). The overexpressed genes in the shared DEGs were classified into 95 functional GO terms, including response to (in)organic substances such as "response to organonitrogen compound (GO: 0010243)," "response to karrikin (GO: 0080167)," "response to inorganic substance (GO: 0010035)," and "response to hormone (GO: 0009725)" and metabolism of secondary metabolites such as "phenylpropanoid metabolic process (GO: 0009698)" and "(−)-E-beta-caryophyllene synthase activity (GO: 0080016)" (Fig. 3C).

Specifically overexpressed genes in APs and SPs were assigned to 14 and 82 terms, respectively (Supplemental Table S3). The differently overrepresented GO terms related to nitrogen response and metabolism ("cellular nitrogen compound metabolic process [GO: 0034641]" in APs and "response to organonitrogen compound [GO: 0010243]" in SPs) (Supplemental Fig. S3, B and C) were consistent with the differences in applied nitrogen form between APs as inorganic matter in the culture medium and SPs as organic matter from the fungal partner. In addition, different GO terms related to secondary metabolism were overrepresented in AP ("cellular aromatic compound metabolic process [GO: 0006725]") and SP ("cellular alcohol metabolic process [GO: 0044107]") (Supplemental Fig. S3, B and C). In the SP, GO terms associated with the transporter activity, such as "intracellular transport (GO: 0046907)" and "efflux transmembrane transporter activity (GO: 0015562)," were specifically overrepresented, probably reflecting nutrient transport between plants and fungi (Supplemental Fig. S3C).

Bioactive GA is converted to the inactive form during *B. striata* seed germination

To understand the intrinsic metabolic and signaling of GA during seed germination and fungal colonization, we performed the Kyoto Encyclopedia of Genes and Genomes analysis on the common DEGs, termed as "response to hormone (GO: 0009725)." This analysis revealed that the ent-kaurene oxidase gene (TRINITY_DN56117_c0_g1_i2_g.199526_m.199526), which is involved in the GA synthesis, and the *DELLA* gene (TRINITY_DN29017_c0_g1_i1_g.90488_m.90488), which is a negative regulator of GA signaling, were detected as commonly upregulated genes in APs and SPs (Supplemental Fig. S4). This result was inconsistent with *Arabidopsis* seed germination, in which *DELLA* genes were either constantly expressed or gradually decreased from 12 h to 2 d after imbibition (Tyler et al. 2004), and the rice *DELLA* ortholog *SLENDER RICE1* expression was almost unchanged during seed germination (Log_2FC of 0.49) (Fig. 4A). In cereal seeds, the α-amylase-encoding gene acts as one of the downstream genes under GA-mediated seed

germination (Gubler et al. 2002). A GA-regulated transcriptional factor, GAMYB, activated α-amylase-encoding gene expression and was promoted by GA-triggered degradation of DELLA protein *SLENDER1* in the barley (*Hordeum vulgare*) aleurone (Gubler et al. 2002). Consistently, the expression of α-amylase genes was markedly increased in germinated rice seeds (Supplemental Fig. S3D). In contrast, there were no significant differences between APs at Week 1 and 0 seeds (Supplemental Fig. S3D). In addition, according to the other transcriptomic studies of symbiotically germinated *Dendrobium officinale* inoculated with *Tulasnella* sp. Pv-PC-2-1 (Wang et al. 2018), common upregulated DEGs at Week 1 included 2 carotenoid cleavage dioxygenase-encoding genes (*CCD7* and *CCD8*), which are necessary for strigolactone (SL) biosynthesis (Fig. 4B). Previous studies reported that the exogenous GA inhibited SL biosynthesis and exudation in *O. sativa*, *L. japonicus*, and the lisianthus (*Eustoma grandiflorum*) roots (Ito et al. 2017; Tominaga et al. 2021). Consistently, both *CCD7* and *CCD8* were upregulated in AM-colonized rice roots (Log_2FC of 1.43 and 1.89, respectively), whereas only *CCD7* was detected as significantly overexpressed in germinated seeds in rice (Fig. 4B).

Given that GA negatively controls both seed germination and fungal colonization, and the APs and SPs shared a large number of DEGs, including GA biosynthesis and signaling genes, it is expected that *B. striata* optimizes GA production during seed germination to coordinate both seed germination and fungal colonization. The transcriptome analysis showed that GA 2-oxidase-encoding genes (*GA2oxs*), encoding enzymes that convert bioactive GAs to inactive forms, and GA 3- and 20-oxidase-encoding genes (*GA3oxs* and *GA20oxs*, respectively), encoding the biosynthesis enzymes to produce the active form of GAs, were significantly upregulated in APs and SPs (Fig. 5A). These significant changes in the expression of *GA2oxs* and *GA20ox* genes were also detected in rice seed germination but moderately found in AM-colonized rice roots (Fig. 5A). To reveal more details of the regulation of GA metabolism, gene expression analysis was performed by reverse transcription quantitative PCR (RT-qPCR). In APs, the average expression of a *BsGA2ox* was upregulated 19 and 45-fold at Weeks 1 and 2, respectively, compared with Week 0, which was the start of the experiment, although the increase during Week 1 was not significant in this experiment (Fig. 5B). Similarly, the average expression of *BsGA3ox* was 12 and 40 times higher at Weeks 1 and 2, respectively, whereas *BsGA20ox* was not significantly altered. In SPs, the expression of *BsGA3ox* and *BsGA20ox* was significantly upregulated at Weeks 2 and 1, respectively, and *BsGA2ox* expression was more than 100 times higher during the period (Fig. 5C).

To further confirm these findings, we conducted a quantitative analysis of endogenous GA in APs and SPs at Week 2 and 0 seeds using LC-MS/MS. The study showed endogenous bioactive GA precursors (GA_{19} and GA_{53}) and an inactivated GA (GA_8) in the 13-hydroxylation pathway (Fig. 6A), while bioactive GA (GA_1) could not be detected due to the detection limit of the instrument. In APs, the amount of GA_{19} was significantly increased (5.6 times), and the GA_8 was 3.2 times higher but not significant compared with that in seeds (Fig. 6B). The GA_{53} was slightly higher (1.5 times) in APs than in seeds but not significant (Fig. 6B). The quantification analysis also showed that in SPs, the amount of GA_8 and GA_{19} was significantly increased compared with the seeds, whereas no significant difference was observed in the amount of GA_{53} accumulated during the experimental period (Fig. 6B). The gene expression and GA quantification results revealed that the biosynthetic pathway from the GA precursor GA_{19} to the inactive form GA_8 was activated during both seed germination and fungal colonization. Specifically, BsGA2ox, which was strongly expressed in SPs, actively converted bioactive GAs to inactive forms during symbiotic germination. Taken together, these results indicate that although GA biosynthesis is activated during seed germination in *B. striata* and other plant species, such as *Arabidopsis*, rice, and barley (Kaneko et al. 2002; Gubler et al. 2008; Dekkers et al. 2013; Urbanova and Leubner-Metzger 2018), simultaneously, the bioactive GA is actively converted to the inactive form in *B. striata*. In addition, other factors, such as DELLA proteins, could inhibit GA signaling during *B. striata* seed germination.

Asymbiotic germination induced the expression of OM marker genes

Our transcriptome results of shared gene expression patterns between APs and SPs support the notion that the artificial media for asymbiotic germination would mimic something in the field probably provided by fungi (Rasmussen 1992; Eriksson and Kainulainen 2011). In other words, the present findings have led to the hypothesis that symbiotic machinery is activated automatically during seed germination even without fungi. To determine whether the symbiosis-signaling pathway is activated during asymbiotic germination, the expression levels of rice AM colonization marker homologs (Güimil et al. 2005; Gutjahr et al. 2008), part of which were identified as OM markers in *B. striata* (Miura et al. 2018), and the mycorrhiza-specific phosphate transporter gene *PHOSPHATE TRANSPORTER 11* (*PT11*) were compared between rice and *B. striata*. The previous rice transcriptome data showed 9 significantly and highly expressed marker genes in AM-forming roots and only 2 in germinated seeds (Fig. 7). Our time-course transcriptome analysis of *B. striata* revealed that the expression of 7 AM marker homologs was significantly increased in both SPs and APs (Fig. 7). The *PT11* expression was also significantly induced in both APs and SPs as well as AM-forming roots but not in rice-germinated seeds (Fig. 7). Consistently, RT-qPCR analysis of AP showed that the expression of 6 of 8 OM marker genes, including *BsAM2*, the induction of which was not detected by RNA-seq analysis, was upregulated during germination despite the absence of fungus, whereas *BsAM34* was not significant (Fig. 8).

Discussion

It is generally accepted that seed dormancy and germination are determined by the interactive effects between different

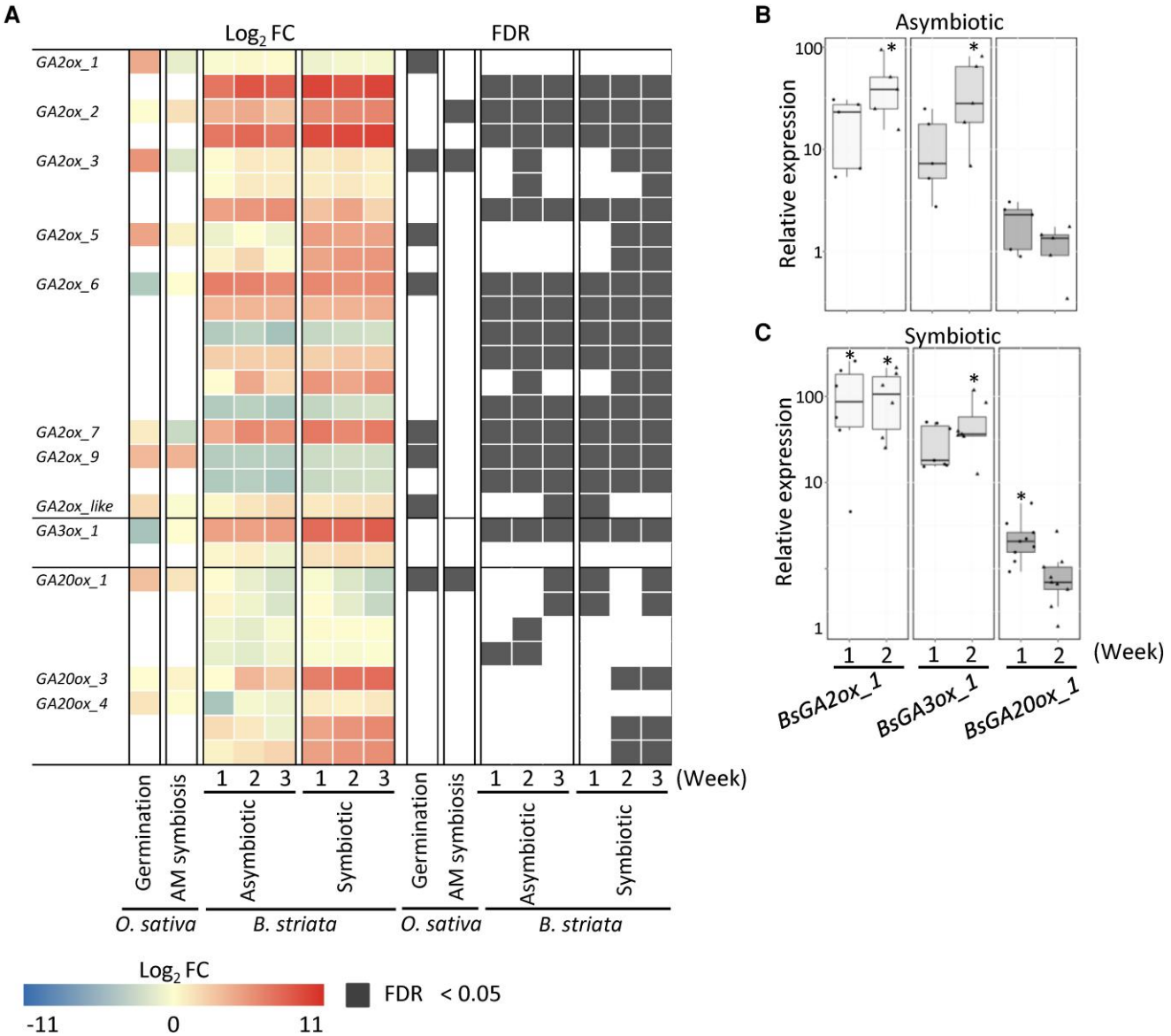

Figure 5. Expression of GA biosynthesis and metabolism genes during asymbiotic and symbiotic germination. **A)** The expression patterns of GA biosynthesis and metabolism genes from RNA-seq data. The heatmap on the left shows the expression patterns of the selected GA-related genes on the basis of Log_2FC. Log_2FC was calculated between time points; 0-wk seeds versus 1 to 3-wk protocorms. "Germination" and "AM symbiosis" indicate germinated seeds and arbuscular mycorrhizal roots of rice, respectively. The right panel displays FDRs. **B, C)** RT-qPCR of GA biosynthesis and metabolism genes. Relative expression analysis was performed with total RNA isolated from asymbiotically **B)** or symbiotically **C)** germinated *B. striata* at different time points (Week 0 seeds to protocorms at 1 or 2 wk after sowing [WAS]). The relative expression values were determined using the relative standard curve method. The FC in expression is relative to 0-wk-old seeds (expression level = 1). Data are shown as box plots with the heavy line in the box representing the median (50th percentile), the ends of the box representing 25th and 75th percentiles, respectively, and the whiskers showing the smallest and largest value. Black dots and triangles indicate the individual data points. The *y*-axis scale is logarithmic. Asterisks indicate significant differences compared with the 0-wk-old seeds using a Bonferroni-adjusted pairwise *t*-test ($n = 5$ to 9, $^*P < 0.05$).

phytohormones such as ABA and GA and that GA is necessary for seed germination (Miransari and Smith 2014). However, our results show that exogenous GA, as well as ABA, significantly inhibits the germination of *B. striata* seeds with undifferentiated embryos, even on a filter paper without artificial nutrition. These indicate the existence of a specific mechanism of GA signaling in *B. striata* seed germination (Fig. 9). This is consistent with some earlier studies of terrestrial and epiphytic orchids, including some subfamilies, which reported that GA had a negative or no effect during germination (Hadley and Harvais 1968; Van Waes and Debergh 1986; Wilkinson et al. 1994; Miyoshi and Mii 1995; Chen et al. 2020). These results imply that orchid species broadly conserve this unique GA signaling pathway. Figura et al. (2019) found that

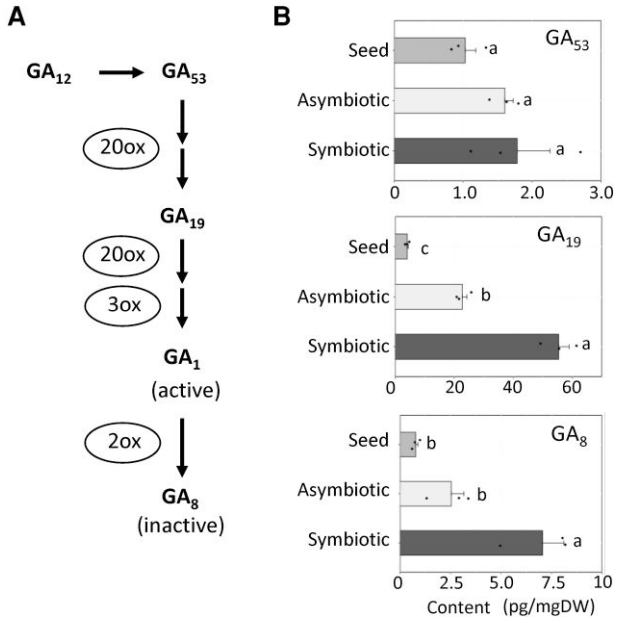

Figure 6. Quantification of endogenous GAs in *B. striata* protocorms. **A)** Simplified metabolic scheme of GA biosynthesis. The "20ox," "3ox," and "2ox" mean GA 20-, 3-, and 2-oxidase enzymes, respectively. **B)** The content of endogenous GAs during seed germination. The GAs were detected in seeds at Week 0 ("seed"), APs and SPs at Week 2 ("asymbiotic" and "symbiotic," respectively). Different letters indicate statistically significant differences on the basis of the Bonferroni-adjusted pairwise *t*-test ($n = 3$, $P < 0.05$). Each bar represents the mean value ± SE.

GA had little effect on seed germination of some initially mycoheterotrophic Pyroloideae (Ericaceae) plants possessing tiny seeds with a 1-layered endosperm. Conversely, root-parasitic plants belonging to the Orobanchaceae also produce tiny seeds where GA acts as a positive regulator in their seed germination in response to SLs (Zehhar et al. 2002; Uematsu et al. 2007). Further studies are required to understand the association between GA sensitivity and heterotrophic evolution.

Connecting multiple signaling pathways has the potential to acquire new adaptive functions (True and Carroll 2002). Previous studies on model plants have reported that the gene regulatory mechanisms stimulated by circadian and environmental cues contribute to the resistance against a plant pathogen (Wang et al. 2011) and the facilitation of AM symbiosis (Umehara et al. 2008; Balzergue et al. 2011; Kretzschmar et al. 2012). These studies have established the concept of molecular links between intrinsic or extrinsic cues as an input and signaling pathways that seem far apart from these cues as an output. Our results showed that GA has lost its role as an activator of α-amylase gene expression during orchid seed germination. Instead, an exogenous germination stimulator probably derived from OM fungi, but not the fungal colonization itself, activated symbiosis-related genes shared with AM symbiosis through GA inactivation, leading to simultaneous control of seed germination and fungal colonization. In the AM symbiosis signaling mutant *ccamk-1* of rice wherein the mutation blocked AM colonization at the root epidermis, no increase in AM marker expression, except for *OsAM1* and *OsAM2*, was observed in AM-inoculated roots (Gutjahr et al. 2008). The enhancement of the expression of *B. striata* OM marker genes in APs could result from the activation of symbiosis signaling by germination stimulators not by fungal colonization. These results suggest that orchids autoactivate the mycorrhizal symbiosis pathway through GA inactivation during seed germination to accept the fungal partner immediately. In addition, the results showing that the number of DEGs identified from SP transcripts was higher than that from AP, one of the OM marker genes *BsAM39* was expressed only in SP, and the *GA2ox* expression level was more stimulated in SPs than in AP suggest that the OM symbiosis pathway is activated in each of the 2 steps of symbiotic germination: the initiation of seed germination and fungal colonization.

In contrast to the exogenous GA effect, the GA-biosynthesis inhibitor uniconazole-P positively regulates *B. striata* seed germination, although the germination rate decreased at high concentrations, probably because uniconazole-P affects not only GA biosynthesis but also the metabolism of some phytohormones (Izumi et al. 1988; Iwasaki and Shibaok 1991). Previously, it was thought that orchid seeds need to be colonized by fungi to germinate even in vitro (Arditti 1967). However, since Lewis Knudson in 1922 formulated the asymbiotic germination method, the fact that germination occurs when seeds are placed in a relevant medium containing appropriate substances, such as sugars, mineral salts, vitamins, amino acids, and phytohormones without fungus, is broadly accepted (Arditti 1967). Thus, artificial media for asymbiotic germination would mimic something in the field probably provided by fungi (Rasmussen 1992; Eriksson and Kainulainen 2011). Our results may support the hypothesis that orchid germination is stimulated by environmental factors instead of fungal infection because exogenously treated uniconazole-P stimulated *B. striata* germination even without any other nutrients and fungi. Moreover, the uniconazole-P acts as a germination stimulator commercially accessible and usable by gardeners, offering fundamental knowledge to the development of conservation and restoration methods of orchids facing extinction (Swarts and Dixon 2009).

Although the actual influencing environmental factors are still unknown, our results suggest that the unknown factors induce the GA metabolic pathway, resulting in the stimulation of seed germination and fungal colonization. In both APs and SPs, the expression of *GA2ox* was significantly increased. Consistently, the inactivated GA, GA_8, accumulates in APs and SPs but is not statistically different between APs and seeds before sowing. A previous transcriptomic study has also reported high expression of genes related to the GA-GIBBERELLIN INSENSITIVE DWARF1-DELLA signaling module, including *GA2ox* and *GA20ox*, in the protocorms of the Jewel orchid (*Anoectochilus roxburghii*) inoculated with unknown fungal species (Liu et al. 2015). These results

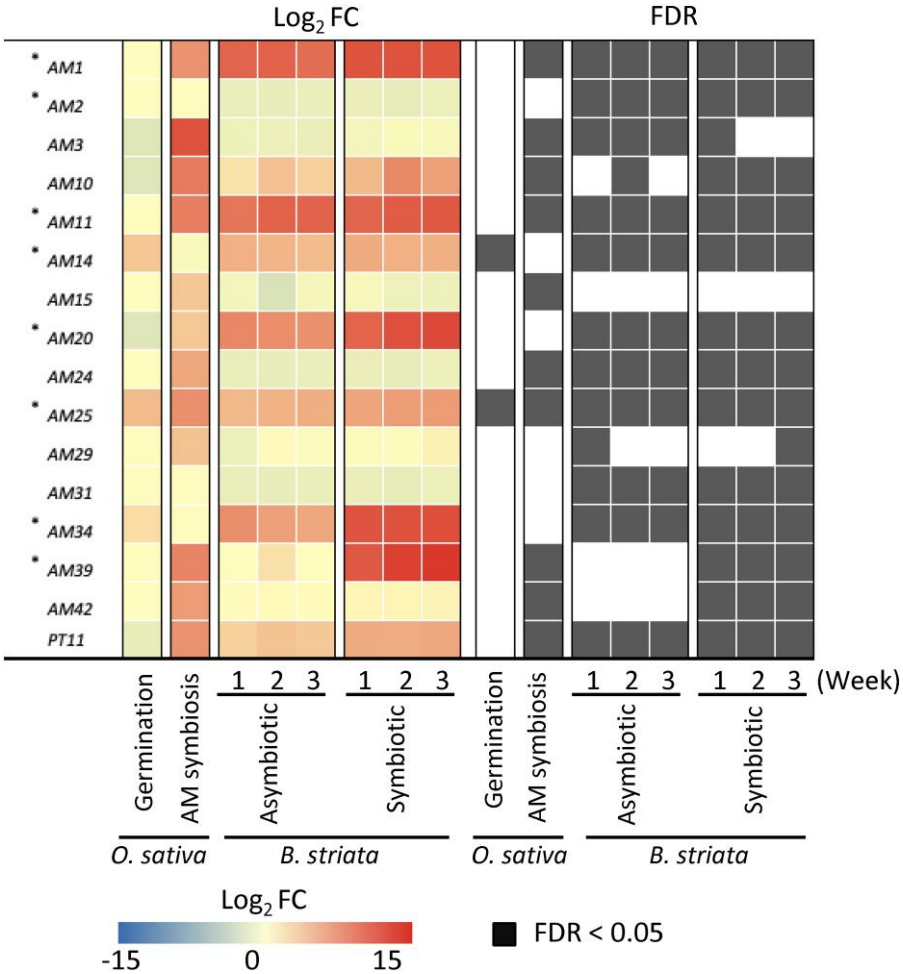

Figure 7. The expression patterns of symbiosis marker genes of rice and *B. striata* from RNA-seq data. The heatmap on the left indicates the expression patterns of the arbuscular mycorrhizal symbiosis marker genes (Gutjahr et al. 2008) and the mycorrhiza-specific phosphate transporter *PT11* gene on the basis of Log_2FC. In rice, Log_2FC was calculated between Day 0 seeds and germinated seeds at 2 d after seeding ("germination") and between noncolonized roots and AM roots ("AM symbiosis"). In *B. striata*, Log_2FC was computed for Week 0 seeds versus 1- to 3-wk protocorms. The right panel displays FDRs. Asterisks indicate the OM symbiosis marker genes (Miura et al. 2018).

indicate that the bioactive GA is actively converted to the inactive form during both seed germination and fungal colonization in orchids. The negative effect of bioactive GA in symbiotic germination resembles the results of earlier studies on AM symbiosis, which reported that GA signaling negatively affects AM fungal colonization and development (Floss et al. 2013; Foo et al. 2013; Takeda et al. 2015). Our previous study found that the key components of AM symbiosis and their concomitant expression pattern are shared in OM symbiosis (Miura et al. 2018). Given that the fungal partner is necessary for orchid seed germination under natural conditions, these findings suggest that the GA is converted from the active form to the inactive form to establish and maintain symbiotic associations during seed germination.

Orchid seeds containing almost no or few resources are tiny and require the presence of compatible fungi to obtain nutrients, resulting in mycoheterotrophy (Merckx 2013). The evolutionary process leading to the formation of dust-like seeds in such mycoheterotrophic plants is supposed to be complex because many drivers behind fungal specificity, chemical interaction, metabolite transport, and plastid genome evolution are responsible for this unique plant–fungus relationship (Eriksson and Kainulainen 2011; Merckx 2013). Rasmussen and Rasmussen (2014) suggest that the evolution of seedling parasitism precedes embryo size reduction and loss of endosperm. Our findings imply that orchids have co-opted the AM symbiotic signaling pathway in roots for GA-mediated germination to coordinate these 2 concurrent events. These molecular links between plant development and symbiosis signaling will provide clues to the mechanisms of the evolution of mycoheterotrophic germination.

Materials and methods

Plant materials and fungal strain

Information on plant species used in this study was listed in Supplemental Table S4. Seeds of a hyacinth orchid (*B. striata*

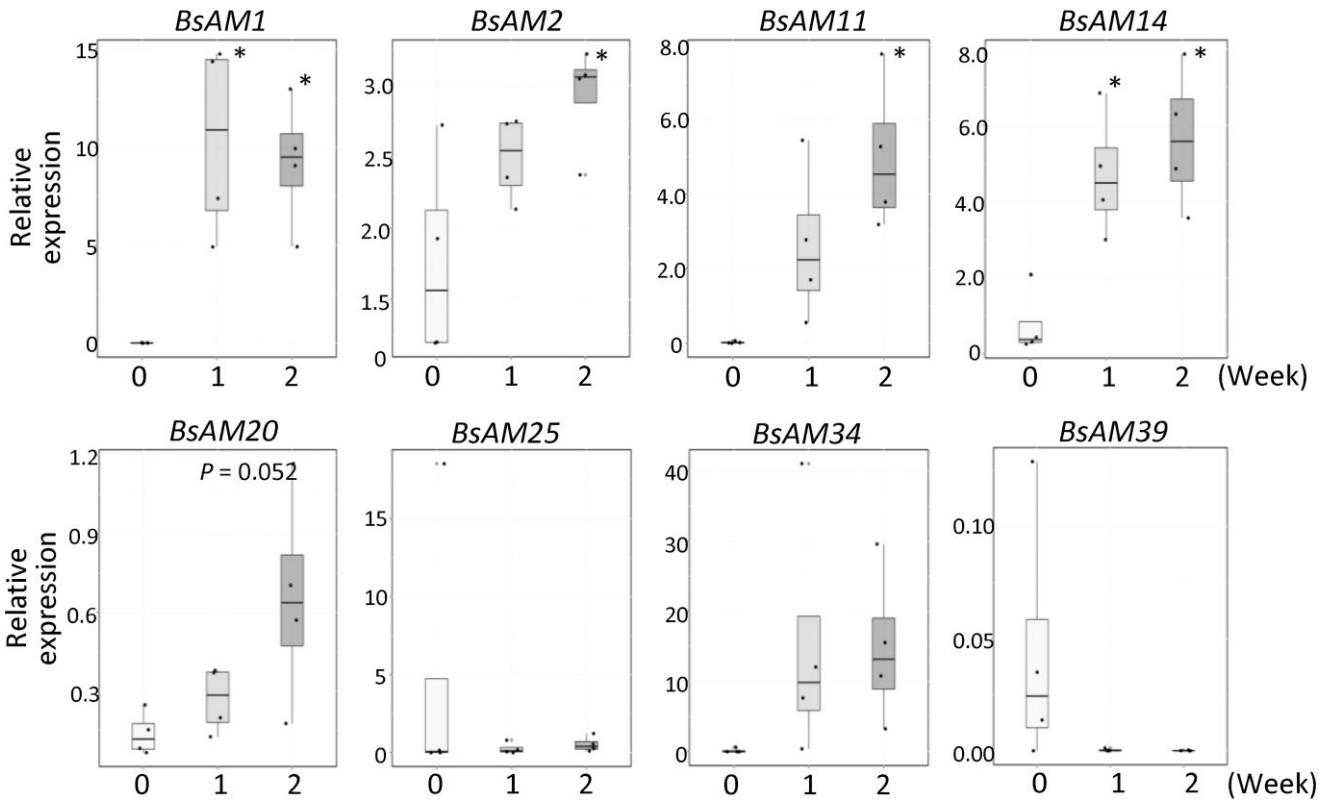

Figure 8. RT-qPCR of OM symbiosis marker genes. Relative expression analysis was performed with total RNA isolated at different time points (Week 0 seeds to germinated protocorms at 2 wk after seeding [WAS]). The relative expression values were determined using the relative standard curve method. Data are shown as box plots with the heavy line in the box representing the median (50th percentile), the ends of the box representing 25th and 75th percentiles, respectively, and the whiskers showing the smallest and largest value. Black and gray dots indicate the individual data points and outliers, respectively. Asterisks indicate significant differences compared with the 0-wk-old seeds using the Bonferroni-adjusted pairwise *t*-test ($n = 4$, $^*P < 0.05$).

cv. Murasaki Shikibu) and its symbiotic fungus, *Tulasnella* sp. strain HR1-1, were mainly used in this work. The origins of the plant and fungal line have been described in detail previously (Yamamoto et al. 2017). Whole plants of *S. australis* and *C. appendiculata* var. *variabilis* were collected from natural habitats in Japan and grown in a greenhouse. Mature seeds of *B. striata*, *S. australis*, and *C. appendiculata* var. *variabilis* were harvested at 5, 1, or 4 mo after manual self-pollination, respectively. Mature seeds of *G. crassifolia* and *V. falcata* cv. Beniohgi were collected from natural habitats in Japan. Collected seeds were stored at 4 °C until they were used. Fungal colonies were cultivated on potato dextrose agar (PDA; Kyokuto, Tokyo, Japan) medium at 25 °C before the symbiotic germination experiments.

Asymbiotic and symbiotic germination on agar medium

Asymbiotic germination procedures were performed according to the method of Yamamoto et al. (2017), either with or without additional growth regulators: gibberellic acid 3 (GA_3) (Nacalai Tesque, Kyoto, Japan), uniconazole-P (FUJIFILM Wako Pure Chemical, Kyoto, Japan), and ABA (Sigma-Aldrich, St. Louis, MO, USA) dissolved in 100% (v/v) ethanol, and Sumiseven P (Sumitomo Chemical, Tokyo, Japan) containing 10-ppb uniconazole-P and Bounty Flowable (Syngenta Japan, Tokyo, Japan) containing 10-ppb paclobutrazol dissolved in distilled water. The method is briefly described below. The seeds were surface sterilized in 1% (v/v) sodium hypochlorite solution for 2 min for *B. striata* and *V. falcata*, 30 s for *S. australis*, 4 min for *C. appendiculata*, and 5 min for *G. crassifolia*. Sterilization time was considered by observing seeds soaked in sterile solution for 20 min when the preliminary experiment was conducted. It was long enough to prevent microbial contamination before the seed coat disintegration and embryo discoloring. After rinsing with sterilized distilled water, 40 to 800 sterilized seeds per plate were sown onto HA medium (Lee and Yeung 2018) (3.0-g Hyponex [N:P:K = 6.5:6:19] [Hyponex Japan, Osaka, Japan], 2.0-g peptone, 30-g sucrose, 10-g agar, 1-L distilled water, and pH 5.5). Symbiotic germination of *B. striata* was performed according to the method of Yamamoto et al. (2017). Sterilized seeds were sown on OMA medium (Lee and Yeung 2018) (2.5-g OMA [Difco, Franklin, NJ, USA], 6.5-g agar, 1-L distilled water, and pH 5.5), which was preinoculated with a culture of *Tulasnella* sp. for a week at 25 °C in the dark. Growth regulators were added to the

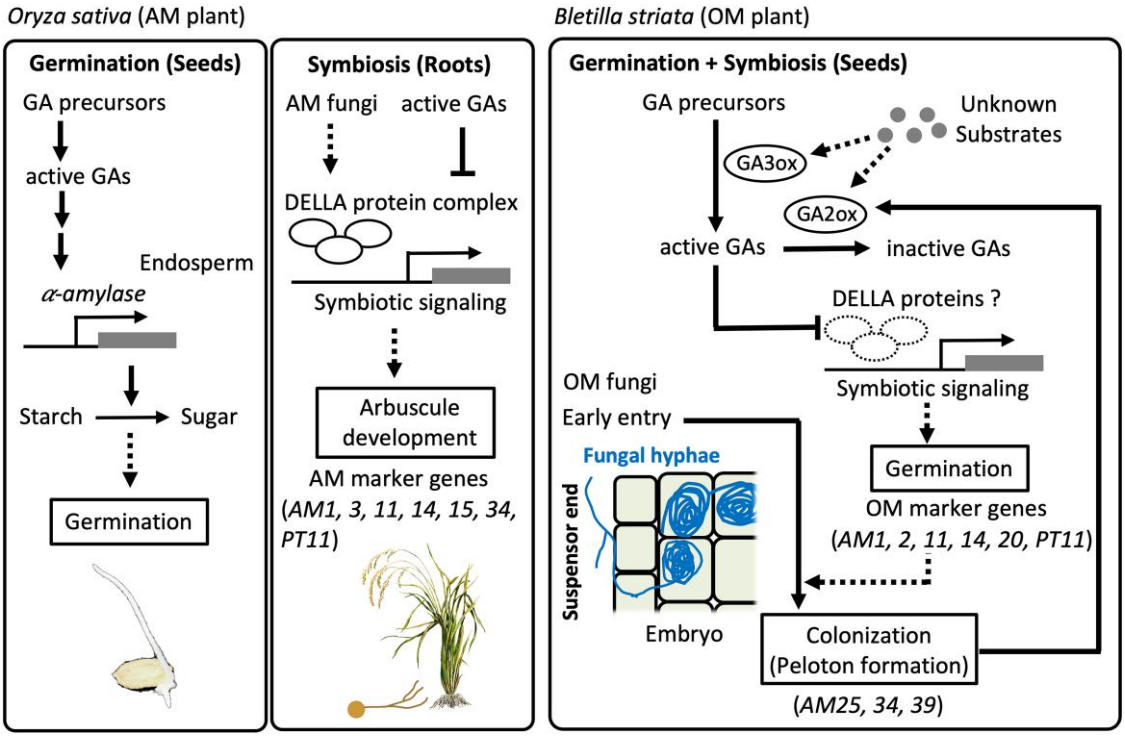

Figure 9. Proposed model for seed germination mechanism in orchids. GA stimulates seed germination of AM plants, such as *O. sativa*, by inducing the expression of α-amylases necessary for the utilization of starch stored in the endosperm (Kaneko et al. 2002). After root development, the mutual relationships between plants and AM fungi are established in the roots. Exogenous-treated GA inhibits fungal colonization in rice through the degradation of DELLA proteins. In *B. striata*, an OM plant, exogenous treatment with GA inhibits seed germination and fungal colonization via unknown mechanisms. When seed germination occurs, environmental factors probably derived from OM fungi stimulate the expression of GA metabolic genes such as GA3ox and GA2ox, leading to symbiotic signaling even without fungi. OM fungi form pelotons in the cortical layer, which promotes *GA2ox* gene expression.

autoclaved media after cooling to 60 °C. Ethanol (0.01%, v/v) was included in the media as a control treatment. The germination experiments were conducted at 25 °C in the dark. The seeds cultured on the media were observed to calculate the germination rate or harvested and stored in a formalin–acetic acid–alcohol (FAA) solution at 4 °C for cell staining or at −80 °C after freezing in liquid nitrogen for RNA extraction and phytohormone quantification. Fungal growth area was measured after 7 d using the ImageJ software (https://imagej.net/ij/index.html).

Quantitative measurement of seed germination and symbiotic cells

The germination was defined as the emergence of a shoot in *B. striata* and *V. falcata* or rhizoids in *S. australis*, *G. crassifolia*, and *C. appendiculata*. To measure the germination rate, all of the seeds sown on 3 to 5 replicate plates per treatment were observed under a stereomicroscope (Olympus SZX16, Tokyo, Japan). The experiments were independently performed 3 times for asymbiotic or symbiotic germination of *B. striata*. Asymbiotic germination experiments of *S. australis*, *G. crassifolia*, and *V. falcata* were performed independently 3 times. For *C. appendiculata*, seed germination experiments were performed only twice due to the small number of seeds available in this study. The FAA-fixed protocorms were rinsed with distilled water and softened in 10% (w/v) KOH at 105 °C for 5 min. The alkalized samples were rinsed with distilled water, neutralized by 2% (v/v) HCl for 5 min, and stained with 10% (v/v) ink dye solution (10% [v/v] Pelikan 4001 Brilliant Black and 3% [v/v] acetic acid) at 95 °C for 30 min for counting of symbiotic cells or with 10-μg/ml WGA-Alexa Fluor 488 solution (Thermo Fisher Scientific, Waltham, MA, USA) overnight and 1-μg/ml calcofluor white solution (Sigma-Aldrich, St Louis, MO, USA) for 15 min in the dark at room temperature for observation of the fungal distribution. Ink-stained materials were soaked in lactic acid (Nacalai Tesque, Kyoto, Japan) at 4 °C before microscopic observation. Procedures of symbiotic cell counting were performed according to the method of Yamamoto et al. (2017). The method is briefly described below. The protocorms with their seed coat removed were lightly crushed between a glass slide and a cover glass. The number of all symbiotic cells in a protocorm was counted under a BX53 light microscope (Olympus) in 5, randomly selected protocorms of each treatment. Results were averaged per treatment. The cell counting experiment was independently repeated 4

times per treatment. The fluorescent images were observed under a DM2500 microscope (Leica Microsystems, Wetzlar, Germany) with filter cube L5 for WGA-Alexa Fluor 488 and filter cube A for calcofluor white.

Asymbiotic germination on filter paper

Sterilized *B. striata* seeds were placed on a filter paper (ø 70 mm) containing 2 ml of test solutions, which included 1 μM GA_3 or 1 μM uniconazole-P. Ethanol (0.01%, v/v) was used as a control treatment. The seeds of rice (*O. sativa*) cv. Nipponbare and *L. japonicus* MG-20 were also seeded on a filter paper using the same procedure. These seeds were incubated at 25 °C for 3 wk, 3 d, and 2 d in the dark for *B. striata*, *O. sativa*, and *L. japonicus*, respectively. Germination was defined as the emergence of a root in *O. sativa* and *L. japonicus*. The experiment was independently performed 3 times with 3 to 5 replicate plates for each treatment.

RNA preparation and gene expression analysis by RT-qPCR

Total RNA was extracted from approximately 5 mg of pooled *B. striata* seeds or protocorms for each condition using the Total RNA Extraction Kit Mini for Plants (RBC Bioscience, New Taipei, Taiwan), following the manufacturer's protocol. First-strand cDNA synthesis was performed using the ReverTra Ace qPCR RT Master Mix with gDNA Remover (Toyobo, Osaka, Japan) following the manufacturer's protocol. RT-qPCR assays were carried out using the THUNDERBIRD SYBR qPCR Mix (Toyobo) on a CFX connect real-time detection system (Bio-Rad Laboratories, Hercules, CA, USA) using the following program: 95 °C for 10 min; 45 cycles of 95 °C for 30 s, 60 °C for 30 s, and 72 °C for 30 s; and a final extension at 72 °C for 5 min. FCs were calculated using the expression of a housekeeping gene, *UBIQUITIN5*, in *B. striata* as the internal control. The gene-specific primer sequences used in this study are shown in Supplemental Table S5 and Miura et al. (2018). The cDNA synthesized from approximately 100-ng total RNA was regarded as 1 biological replicate. At least 4 biological replicates from independent experiments containing 3 technical replicates each were performed. After the threshold cycle (*Ct*) values were averaged, the FCs were calculated using the delta–delta cycle threshold method (Livak and Schmittgen 2001) or the relative standard curve method (Pfaffl 2001).

Quantification of endogenous GA levels in *B. striata*

Asymbiotically or symbiotically germinated *B. striata* protocorms growing on the medium without phytohormone treatments were harvested 2 wk after sowing and stored at −80 °C after freezing in liquid nitrogen. The seeds, before sowing, were also stored at −80 °C as a control sample. The frozen samples were weighed after lyophilization. After grounding and homogenizing, the 35.15- to 234.89-mg samples were subjected to LC–MS/MS to quantify the endogenous GAs according to the method of Kanno et al. (2016). Data were obtained from 3 independent replicate experiments.

RNA-seq and data analysis

The total RNA of Week 0 seeds as a control sample and APs was treated with RNase-free DNase I to remove residual genomic DNA and cleaned using the RNeasy Mini Kit (Qiagen, Hilden, Germany) according to the manufacturer's protocol. The quality and quantity of the purified RNA were confirmed by measuring its absorbance at 260 and 280 nm (A260/A280) using a NanoDrop ND-1000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA) and by electrophoresis using an Agilent 2100 Bioanalyzer (Agilent Technologies, Santa Clara, CA, USA). RNA-seq library was constructed from the total extracted RNA using the Illumina TruSeq RNA Library Prep Kit (Illumina, San Diego, CA, USA) according to the manufacturer's protocol. Multiple cDNA libraries were sequenced using the Illumina HiSeq platform with 100- or 125-bp single-end reads. The seeds or protocorms collected from 5 different plates were pooled as 1 biological replicate. Three biological replicates were prepared per conditions for the transcriptome analysis. Consequently, more than 7 million raw reads per sample were obtained (Supplemental Table S3). Low-quality reads (<Q30) and adapter sequences were filtered and trimmed using Fastp (Chen et al. 2018).

Differential expression analysis

For data analysis, raw reads of seeds before germination and asymbiotically geminated *B. striata* generated in this study (the DNA Data Bank of Japan [DDBJ] DRA accession DRR439921–DRR439932) and symbiotically germinated *B. striata* obtained from our previous study (DDBJ DRA accession DRR099058–DRR099075) (Miura et al. 2018) were analyzed. The reads were mapped using Bowtie 2 (Langmead and Salzberg 2012), and the transcript abundance was estimated using eXpress ver. 1.5.1 (Roberts and Pachter 2012) in accordance with the method of Miura et al. (2018). Differences in library size were corrected using the trimmed mean of *M*-value normalization method, and EdgeR (Robinson et al. 2010) was used to identify DEGs with a Log_2FC of $\geq$1.0 or $\leq$−1.0 and FDRs of <0.05. The time-course data were analyzed using a general linear model. To identify the DEGs in germinated and AM-colonized rice, publicly available short-read data were obtained from the Short-Read Archives at the National Center for Biotechnology Information BioProject accession PRJNA474721 (Narsai et al. 2017) and DDBJ BioProject accession PRJDB4933 (Kobae et al. 2018), respectively. The downloaded raw reads were filtered and trimmed using Fastp with the option "-q 30" (Chen et al. 2018) and then mapped using STAR (Dobin et al. 2013). The reference genome (Oryza_sativa IRGSP-1.0.48) was downloaded from Ensembl plants (https://plants.ensembl.org/index.html). The aligned reads were counted using featureCounts (Liao et al. 2014); subsequently, the genes were subjected to differential expression analysis using the EdgeR

package (Robinson et al. 2010). DEGs were defined as Log_2 FC of ≥1.0 or ≤−1.0 and FDR of <0.05.

Gene annotation and GO analysis

The *B. striata* de novo reference assembly (Miura et al. 2018) was functionally annotated with EnTAP (Hart et al. 2020) using the NCBI NR, Plant RefSeq, and UniProt (https://www.uniprot.org/) databases. The GO enrichment analysis of DEGs was performed using the topGO package in the R environment.

Statistical analyses

All statistical analyses were conducted using R software v4.2.3. To determine the significance of any difference in a number of symbiotic cells, germination rate of APs, and relative gene expression levels, the normality of the data was assessed by Shapiro–Wilk test followed by Welch's *t*-test corrected by the Bonferroni method ($\alpha = 0.05$). The differences in the germination rates of SPs were checked using Student's *t*-test ($\alpha = 0.05$). The elim methods in the case of the Kolmogorov–Smirnov test were used to detect overrepresented GO terms within 3 ontologies: "biological process," "cellular component," and "molecular function" ($P < 0.01$).

Accession numbers

Sequence data from this article can be found in the GenBank/EMBL data libraries under the accession number DRA015645 (BioProject PRJDB14881).

Acknowledgments

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Author contributions

C.M. and H.K. conceived and designed the analysis. C.M., Y.F., T.Y., M.H., K.S., T.Y., and M.Y. contributed to sample preparation. C.M., Y.F., T.Y., Y.K., M.H., and K.Y. performed the experiments and analyzed the sequencing data. K.S., T.Y., M.S., S.S., and M.Y. helped supervise the project and contributed to the interpretation of the results. C.M. and H.K. wrote the paper. H.K. supervised the project. All authors approved the final manuscript.

Supplemental data

The following materials are available in the online version of this article.

Supplemental Figure S1. The effect of phytohormones on fungal growth and seed germination.

Supplemental Figure S2. The effect of GA and its inhibitor (uniconazole-P) on seed germination of *O. sativa* and *L. japonicus*.

Supplemental Figure S3. Transcriptome analysis of *O. sativa* and *B. striata*.

Supplemental Figure S4. The Kyoto Encyclopedia of Genes and Genomes analysis of shared overexpressed genes between asymbiotically and symbiotically germinated *B. striata* at Week 1.

Supplemental Table S1. RNA-seq summary of *B. striata*.

Supplemental Table S2. Lists of DEGs.

Supplemental Table S3. Lists of overrepresented GO terms.

Supplemental Table S4. List of plant materials used in this study.

Supplemental Table S5. List of primers used in this study.

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Conflict of interest statement. None declared.

Data availability

Nucleotide sequence data from the RNA-seq analysis in this study have been deposited in the DDBJ BioProject under the accession number PRJDB14881. Correspondence and requests for materials should be addressed to H.K. (kaminaka@tottori-u.ac.jp).

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